

TrES-5b

R-2350

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_140820 · created 2026-08-19T01:14:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456889.9289 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.042
Transit depth (Rp/Rs)^2	3.12 +/- 1.5 [%]
Orbital Inclination (inc)	81.21 +/- 2.98
Airmass coefficient 1 (a1)	0.997 +/- 0.024
Airmass coefficient 2 (a2)	0.0012 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	2.43 %
Best Comparison Star	#3 - [128, 269]
Optimal Method	PSF photometry
Transit Duration (day)	0.044 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.177 ± 0.042 vs 0.142 (deviation 0.67σ)
Duration fit vs published	0.044 d vs 0.0758 d (deviation 0.82σ)
Mid-time O–C	4.2 min
Depth SNR	3.4
Residual scatter	2.43 %

Light curve

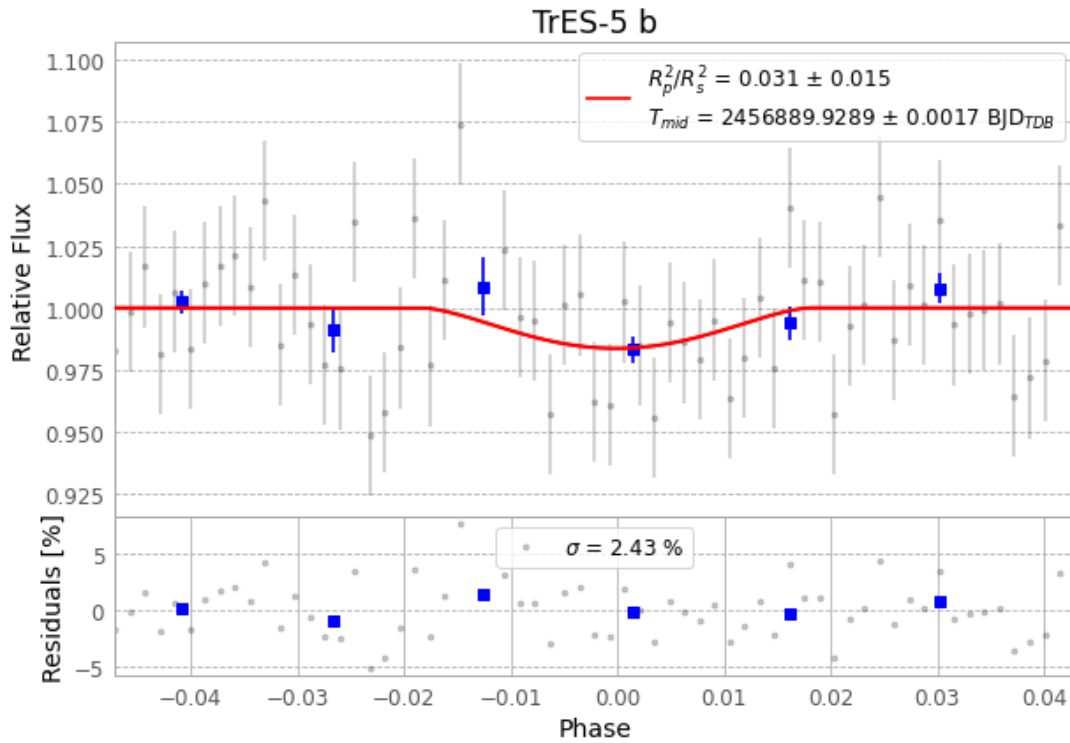


Figure 1 — FinalLightCurve_TrES-5 b_2014-08-20.png (SHA-256 2a11633bdc9f1e2f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d5cb0a97ed1cae6e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

65 FITS frames (43 MB), TRES-5140820083312.FITS → TRES-5140820114513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-140820.log (4084e4d6eba9...), FinalLightCurve_TrES-5 b_2014-08-20.png (2a11633bdc9f...), FinalLightCurve_TrES-5 b_2014-08-20.csv (835e24682e2b...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 01:14Z.